6.2.1 Loftus and Palmer (1974) — Reconstruction of Automobile Destruction

What this workbook covers 本册讲什么

One question, five different verbs, and speed estimates that swing by almost **9 mph**. This booklet walks through **both experiments** — the one that found the effect and the one that explained it — plus the evaluation and the 8-mark answer.

How to use this booklet 怎么用

1. **Fill it in as you go** — check the answer key (pp. 19–20) after each page.
2. **Pages 3–7 (both experiments) and 11 (numbers) matter most.** Do them from memory, under time.

Content map 内容地图

Page	Topic	What you must be able to do afterwards
p.2	One word, two answers 引入	Explain why a single verb matters in court
p.3–4	Experiment one：aim, procedure, results	45 students · 5 verbs · estimates 40.8 → 31.8 mph
p.5	Two possible reasons 两种解释	State response bias vs memory change
p.6–8	Experiment two：aim, procedure, results, conclusion	150 students · broken glass · 32% / 14% / 12%
p.9	Reconstructive hypothesis 重构假设	Define it and give the merging-sources explanation
p.10	Side by side 两实验对照	Fill the full comparison table
p.11	Every number 数字收口	Cover the right column and test yourself
p.12–13	Evaluation 评价	Four strengths, three limitations, each with a consequence
p.14	Real life 现实改变	Devlin Committee and the cognitive interview
p.15	Exam skills 答题技巧	IV / DV, command words, sentence frames
p.16–17	Practice A & B 演练	Short answers, then the 8-mark application
p.18	Final check 五分钟默检	Ten answers from memory, no notes
p.19–20	Answer key 答案速查	Mark honestly, then re-write wrong answers in full

Revision tracker 复习打卡

☐ p.2–4 experiment one ☐ p.5 the two reasons ☐ p.6–8 experiment two ☐ p.9–10 hypothesis & comparison ☐ p.11 numbers ☐ p.12–14 evaluation & real life ☐ p.15–17 exam

The whole study in one sentence 一句话总结

Change one verb in the question — smashed, collided, bumped, hit, contacted — and witnesses estimate speeds from **40.8 down to 31.8 mph**; and a week later, **32%** of “smashed” witnesses remembered **broken glass that was never there** (vs 14% and 12%), proving that post-event information **merges into memory itself** — the reconstructive hypothesis.

§1 · One word, two answers

ANSWERS → p.19

The setup

You witness a crash. Two officers ask about it. One asks how fast the cars were going when they _____ into each other; the other asks about cars that _____ each other. Same crash, one word different.

Why this study is the classic one

Number the three reasons 1–3 in the order given:

☐ It is **two experiments**, not one — the first finds the effect, the second explains it. ☐ It is about a **real problem**: convictions can rest on eyewitness memory. ☐ It **changed practice** — police questioning and the cognitive interview.

> **中文理解：为什么一个动词是大事**
> 目击证词是法庭上**最常用的证据之一**。如果换一个词就能让目击者把 30 迈说成 40 迈， 甚至一周后“记得”片子里根本**不存在的碎玻璃**，那么定罪所依据的记忆本身可能就是**被问话塑造出来的**。 这就是为什么这个研究能改变警察问话的方式。

The exam warning

The textbook is explicit: the research involves _____ experiments, so you must be able to describe experiment _____ and experiment _____ in detail.

§2 · Experiment one — aim and procedure

ANSWERS → p.19

The aim（默写）

To investigate whether _____ questions would influence the _____ of the speed of a vehicle among eyewitnesses.

（leading / estimates）

The procedure — fill the gaps

Feature	Detail
Participants	_____ students, split into _____ groups of _____
Materials	_____ short film clips of traffic accidents, each lasting _____ to _____ seconds
Task	After each clip: give an _____ of the accident, then answer a _____
The one difference	Same questionnaire for all, except _____ critical question — the _____ was changed
The five verbs	_____
Controls	Each of the five groups saw the clips in a _____ order
The check	They knew the actual speeds in _____ of the clips: _____, _____, and two at _____ mph

（45 · five · nine / seven · 5 · 30 / account · questionnaire / one · verb / smashed, collided, bumped, hit, contacted / different / four · 20 · 30 · 40）

为什么“知道真实车速”重要

如果不知道真实车速，就无法排除“**是片子本身开得快**”这个解释。有了这四个数据， 他们发现**真实车速并不影响估计值**，才能有把握地说：是**动词**在起作用。

§3 · Experiment one — the results

ANSWERS → p.19

Table 12.1 — fill the means

Verb used	smashed	collided	bumped	hit	contacted
Mean speed (mph)	_____	_____	_____	_____	_____

（40.8 · 39.3 · 38.1 · 34.0 · 31.8（从左到右按动词强度递减））

Draw the bars — sketch the five means below

Three quick reads of the table

1. Which verb produced the fastest estimate? _____
2. Which produced the slowest? _____
3. The gap between them is almost _____ mph.

怎么记这一列数字

记 **40.8 / 31.8** 两端，中间三个顺着**动词的暴力程度**排：smashed（撞碎）> collided（相撞）> bumped（碰）> hit（撞）> contacted（接触）。动词越狠，估得越快。

§4 · Experiment one — two possible reasons

ANSWERS → p.19

A change of word could _____ affect a witness's answer. But a changed number can mean two different things.

Reason	What it claims
1 · response bias（回答偏差）	The participant was _____ of the speed, and the verb in the leading question created a _____ which influenced their decision.
2 · the memory itself changed	The wording of the question causes a _____ in memory, so they recall the accident as being _____ than it actually was.

（significantly / uncertain · bias / change · more severe）

中文理解：两个理由差在哪

回答偏差：心里知道是 30，顺口说了 40 —— 只是嘴上迎合暗示，**记忆没变**。

记忆改变：脑子里的画面真变成了“猛烈撞击”，**记忆本身被改写**。

只看速度估计，两种解释**分不开** —— 所以才需要实验二。

Why speed estimates alone cannot settle it

Both explanations predict the same pattern in experiment one. To separate them we need a measure that only _____ would produce — something that was _____ in the film at all.

（a changed memory / never present）

§5 · Experiment two — aim and procedure

ANSWERS → p.19

The aim（默写）

To test whether the verb used in the question created a _____ in the participant's response, or whether it created an _____ change in _____ of the incident.

（bias / actual / memory）

The procedure — fill the gaps

Feature	Detail
Participants	_____ students, divided into _____ groups of _____
The film	A multiple car accident lasting _____ one minute; the accident itself lasted _____ seconds
Group 1	“About how fast were the cars going when they _____ into each other?”
Group 2	“…when they _____ each other?”
Group 3	_____ — no critical question about speed at all
The wait	A _____ later, participants returned to answer _____ questions, without watching the film again
Critical question	_____? — Yes or No. There was _____ broken glass in the film.

（150 · three · 50 / less than · 4 / smashed / hit / control / week · ten / “Did you see any broken glass” · no）

三个组各自的作用

smashed 组：听到最暴力的动词。

hit 组：听到中性动词 —— 用来对比。

control 组：**没被问速度** —— 证明“光参加实验”本身不会让人多想出碎玻璃。

一周的延迟是关键：影响不是当天随口应付，而是**留在了记忆里**。

§6 · Experiment two — the results

ANSWERS → p.19

Table 12.2 — fill the counts and percentages

Verb condition	Said “Yes”	Said “No”	% who said “Yes”
“smashed”	_____	_____	_____
“hit”	_____	_____	_____
control	_____	_____	_____

(16 · 34 · 32% / 7 · 43 · 14% / 6 · 44 · 12%)

The statistical test

A _____ test was carried out and there was a _____ difference between the groups: significantly _____ participants in the “smashed” group reported seeing glass than in the “hit” group.

(chi-square / statistically significant / more)

怎么读这三个数字（考试常错）
不要只看“32% 的人说看到”。每组都有**一半以上答对**（说没看到）。关键是**组间差距**： smashed 32% 对 hit 14%，而没人问速度的 control 只有 12%。卡方检验说明差距**不是巧合**。

One-line read

The verb acted as _____ information. It changed their _____ of the event rather than causing a response bias — and it did so even _____ after seeing the film.

(post-event / memory / a week)

§7 · Experiment two — the conclusion

ANSWERS → p.19

Why broken glass settles the argument

A _____ would only move the number people say out loud. It would not make them report a piece of the scene that was _____. Reporting broken glass a week later is evidence that the _____ itself had been rebuilt.

(response bias / never filmed / memory)

The two conclusions（默写）

Loftus and Palmer concluded that _____ memory exists, and that _____ questions do influence eyewitness testimony.

(reconstructive / leading)

The two kinds of information that go into memory

Source	Example in this study
1 · Information from _____ the event	The film clip itself
2 · Information supplied _____ the event	The verb in the critical question

(witnessing / after)

一句话连起来

亲历的信息 + 事后得到的信息 → 随时间**融合（merge）** → 分不清哪条来自哪里 → 只剩**一份合成记忆**。这就是下一页的**重构假设**。

§8 · The reconstructive hypothesis

ANSWERS → p.19

Define it in your own words first

Textbook definition 课本定义

Reconstructive hypothesis: memory for an event when the information supplied **after** the event **merges with** the information obtained from **witnessing** the event.

The chain — fill it in

Witness the event → receive _____ information afterwards → the two sources _____ as time elapses → we cannot tell the _____ of some details → one _____ memory of the event.

（post-event / merge / source / overall）

中文理解：一份"合成的"记忆

生活类比：手机相册里**自己拍的照片**和朋友发来的**截图**混在一起。过一阵子再看， 分不清哪张是自己拍的——但你会**坚信自己当时在场**。

记忆只存**一份合成版本**，不保留"这条来自哪里"的标签。所以目击者未必在说谎， 但**记忆本身可能已经失真**。

Apply it — one exam-style sentence

Write one sentence linking the broken-glass finding to the reconstructive hypothesis:

§9 · Two experiments, side by side

ANSWERS → p.19

	Experiment one	Experiment two
Aim	Do leading questions influence _____ estimates?	Is it a response _____ or an actual _____ change?
Participants	_____ students — five groups of _____	_____ students — three groups of _____
Materials	_____ clips, each 5–30 seconds	_____ film, under a minute (accident 4 seconds)
The verb(s)	_____ verbs: smashed, collided, bumped, hit, contacted	_____ verbs: smashed, hit — plus a _____ with no speed question
The measure	Speed estimate in _____ (quantitative)	_____ to "Did you see any broken glass?"
Timing	Asked _____ after each clip	Asked a _____ later, after ten other questions
Result	40.8 down to _____ mph — a gap of almost _____ mph	_____% vs _____% vs _____%
Proves	A word changes the _____	The _____ itself was changed — not just the answer

(speed / bias · memory / 45 · nine / 150 · 50 / seven · one / five · two · control / mph · Yes/No / immediately · week / 31.8 · 9 / 32 · 14 · 12 / answer · memory)

The strongest describe answers put both side by side

Same idea, different samples, different measures, different weeks. Never write about only one experiment.

§10 · Every number 数字收口

COVER THE RIGHT COLUMN

Question	Answer
Experiment one participants / groups	_____ students, _____ groups of _____
Experiment one clips / length	_____ clips, _____–_____ seconds
“Smashed” mean / “contacted” mean	_____ mph / _____ mph
The gap between the two extreme verbs	almost _____ mph
Actual speeds they knew (four clips)	_____, _____, and two at _____ mph
Experiment two participants / groups	_____ students, _____ groups of _____
Film length / accident length	under _____ minute / _____ seconds
Return after…	a _____, answering _____ questions
Broken glass “Yes”: smashed / hit / control	_____% / _____% / _____%
Broken glass counts: smashed / hit / control	_____ / _____ / _____ said “Yes”
Statistical test used in experiment two	_____
Committee that reviewed eyewitness testimony	_____ (_____)

三组数字的记法

实验一记“45 / 5 组 / 7 段 / 40.8→31.8”。

实验二记“150 / 3 组 / 一周 / 32-14-12”。

顺序：45 在前（找效应）→ 150 在后（解释效应）。

§11 · Evaluating — strengths

ANSWERS → p.20

1 · High control and replicability

Both experiments were conducted in a _____ and therefore had a high level of control. The questions and clips were _____ for the groups. This allows the study to be _____ to determine the _____ of the findings.

(laboratory / consistent / replicated · reliability)

2 · Construct validity

By changing just the _____ associated with speed, they can be more certain that the _____ of the verb was what impacted the estimates. This increases _____ validity — that they tested the effect of _____ on recall.

(verb / IV / construct · language)

3 · Demand characteristics were guarded against

The critical question was randomly _____ among other questions. This prevented participants from _____ the aim of the study and displaying demand characteristics.

(included / guessing)

4 · The real-speed check and quantitative data

They knew the _____ speeds and found it did _____ affect the estimates — so it was the _____, not the film. The yes/no responses and mph estimates gave _____ data with no subjective interpretation, making the findings more _____ and less open to researcher bias.

(actual / did not / verb / quantitative · objective)

四条 strength 的共同点
都在说：**把别的解释一个个堵住了**。写评价时加一句“我们因此能确定是 X 造成的”，比只说“控制得好”得分高。

§12 · Evaluating — limitations

ANSWERS → p.20

1 · Students may have made the effect easier to find

Participants were _____, likely to have _____ experience of car speeds. Limited experience may have made them _____ influenced by the verb than a population with more experience of _____. The findings may only represent this specific _____, rather than generalising to the wider population.

(students · limited · more · driving · population)

2 · Watching a film is not witnessing an accident

Participants were unlikely to be under the same _____ as an eyewitness of a real accident, as they only watched _____ in a laboratory. This makes the _____ questionable — they may have responded differently as a result.

(emotional strain · film clips · validity)

3 · Reductionist

Many other factors could influence memory of a real-life event, including _____ and the _____ in which it occurred. The study is somewhat _____, considering only _____ potential variable.

(emotions · environment · reductionist · one)

三条 limitation 的共同点

都在问：**实验室里的学生，能代表真实目击者吗?**

写 8 分题时，把“学生经验少”和“缺情绪紧张”**分开写**，各配一句后果； 再补一条“只测了一个变量（还原论）”。别笼统写“样本不好”。

§13 · Real life — what changed because of it

ANSWERS → p.20

Devlin Committee (1976)

It was set up to investigate the use of _____ in court. It found that many people have been _____ of serious crimes by eyewitness testimony _____, and recommended against basing a conviction on a _____ eyewitness.

(eyewitness testimony / convicted · alone · single)

What the research led to

Tick what the textbook says:

☐ contributed to the **cognitive interview** ☐ police guided to **minimise leading questions** ☐ a marked change in **policing techniques in the 20th century** ☐ the cognitive interview is still used today ☐ police must hypnotise witnesses

The logic chain（填空）

Question wording can rewrite memory → police must _____ leading questions → the _____ interview was developed → convictions should not rest on a _____ eyewitness (Devlin).

(minimise · cognitive · single)

这一节是 8 分题的“应用”素材

逻辑链：**问话用词会改写记忆** → 警察问话必须**少用引导性问题** → 有了**认知访谈**这套不引导的问法 → Devlin 建议**不能只凭一个目击者定罪**。写“这个研究能不能用到现实”时，这条链直接能用。

Variables — the two short questions

Question	Answer
IV of experiment one	The _____ used in the critical question (smashed, collided, bumped, hit, contacted)
DV of experiment one	The _____ of the cars, in miles per hour
IV of experiment two	The _____ condition: “smashed”, “hit”, or no speed question
DV of experiment two	Whether the participant reported _____ broken glass (Yes / No)

(verb / estimated speed / verb / seeing)

Command words

Command word	What you must do
describe	Both experiments, in order, with the numbers — no evaluation
explain	The mechanism: post-event information merges into memory
assess / evaluate	Claim + evidence + consequence, both sides, then a judgement
apply	Link findings to real eyewitness situations — cognitive interview, Devlin, emotional strain

Sentence frames 句子骨架

- In experiment one, **45 students** watched **seven clips** and estimated speeds; those given “smashed” estimated **40.8 mph** against **31.8 mph** for “contacted”…
- This supports the reconstructive hypothesis because the post-event verb **merged with** the witnessed event…
- However, the sample were **students with limited experience of car speeds**, which may have inflated the effect, so…
- In real life, witnesses are under **emotional strain**, and the study is **reductionist** because it considers only one variable…
- Overall, the study does apply to real eyewitness memory, as shown by the **cognitive interview** and the **Devlin Committee**, but the exact size of the effect may differ…

§15 · Practice A — short answers（先自己写，再对答案）

ANSWERS → p.20

A1. Identify the independent variable and the dependent variable in Loftus and Palmer's (1974) experiment one. (3)

A2. Outline the procedure of experiment two. (4)

提示：150 / 三组各 50 / smashed vs hit vs 无速度问题 / 一周后十个问题 / 碎玻璃问题。

A3. Explain why experiment two was needed after experiment one. (3)

提示：速度估计分不开两种解释；碎玻璃只有“记忆被改写”才能解释。

A4. State one strength and one limitation of this study. (4)

自评标准

每个论点 = 1 个具体名词/数字 + 1 个具体细节。写完先对 p.20 答案，再回来补全句子。

§16 · Practice B — the application question

ANSWERS → p.20

Assess whether Loftus and Palmer's (1974) study can be applied to eyewitness memory in real-life situations. (8)

Plan first — fill the skeleton

For — it does apply（证据 + 数字）	Against — be careful（批评 + 后果）
1.	1.
2.	2.
3.	3.

Judgement（一句话收尾，写在线上）：

评分要点

· AO1：两个实验各带数字（45/150、40.8–31.8、32-14-12）。

· AO3：优点配“所以我们能确定…”；缺点配后果（效应可能被放大 / 生态效度存疑）；补一条还原论。

· 收尾：认知访谈 + Devlin + “有保留地可以应用”。

§17 · Final check — five minutes, from memory

NO NOTES

Rapid recall

1. Experiment one: participants / groups: _____
2. Experiment one: clips and their lengths: _____
3. The five verbs in order of estimate: _____
4. Top and bottom means, and the gap: _____
5. The two possible reasons for the change: _____
6. Experiment two: participants / groups / wait: _____
7. The critical question and the three percentages: _____
8. Reconstructive hypothesis in one sentence: _____
9. Two limitations, each with its consequence: _____
0. Two real-life changes the research contributed to: _____

Brain dump 一句话版本

One verb changes the estimate (40.8 → 31.8 mph), and a week later “smashed” witnesses remember broken glass that was never there (32% vs 14% vs 12%) — post-event information merges into memory itself, so police now avoid leading questions and the cognitive interview exists.

Answer key location

All blanks in this booklet are answered on **pp. 19–20**. Mark each page as you finish it, and re-write anything wrong in full.

p.2 smashed; contacted. Reasons order: 2 (real problem), 3 (changed practice), 1 (two experiments). both; one; two.

p.3 leading; estimates. 45; five; nine / seven; 5; 30 / account; questionnaire / one; verb / smashed, collided, bumped, hit, contacted / different / four; 20; 30; 40.

p.4 40.8 · 39.3 · 38.1 · 34.0 · 31.8. smashed; contacted; 9.

p.5 significantly; uncertain; bias; change; more severe. a changed memory; never present.

p.6 bias; actual; memory. 150; three; 50 / less than; 4 / smashed / hit / control / week; ten / “Did you see any broken glass”; no.

p.7 16 · 34 · 32% / 7 · 43 · 14% / 6 · 44 · 12%. chi-square; statistically significant; more. post-event; memory; a week.

p.8 response bias; never filmed; memory. reconstructive; leading. witnessing; after.

p.9 post-event; merge; source; overall. Model sentence: “Because a week later 32% of the smashed group reported broken glass that was never in the film, the verb must have merged with their memory of the event — supporting the reconstructive hypothesis.”

p.10 speed; bias; memory / 45; nine; 150; 50 / seven; one / five; two; control / mph; Yes–No / immediately; week / 31.8; 9; 32; 14; 12 / answer; memory.

p.11 45 · five · nine / 7 · 5–30 / 40.8 · 31.8 / 9 / 20 · 30 · 40 / 150 · three · 50 / one · 4 / week · ten / 32 · 14 · 12 / 16 · 7 · 6 / chi-square test / Devlin Committee (1976).

p.12 laboratory; consistent; replicated; reliability. verb; IV; construct; language. included; guessing. actual; did not; verb; quantitative; objective.

p.13 students; limited; more; driving; population. emotional strain; film clips; validity. emotions; environment; reductionist; one.

p.14 eyewitness testimony; convicted; alone; single. Ticks: cognitive interview ✓; minimise leading questions ✓; policing techniques in the 20th century ✓; still used today ✓ (hypnotise = NO). minimise; cognitive; single.

p.15 verb / estimated speed / verb / seeing.

p.16 A1 IV = the verb used in the critical question (smashed, collided, bumped, hit, contacted); DV = the estimated speed of the cars in mph. **A2** 150 students in three groups of 50 watched a film of a multiple car accident lasting under a minute (the accident four seconds). One group was asked about cars that “smashed”, one about cars that “hit”, and the control group was not asked about speed. A week later they answered ten questions without seeing the film again, including “Did you see any broken glass?” **A3** Speed estimates alone cannot separate the two explanations. The broken glass was never in the film, so only a changed memory — not a response bias — could make people report it. **A4** Strength: high control, so replicable and reliable. Limitation: students with limited experience of car speeds may be more influenced by the verb, so findings may not generalise.

p.17 B — model judgement “On balance the study can be applied to real eyewitness memory: the effect was strong and lasted a week, the real speeds were controlled for, and its applications — the cognitive interview and the Devlin Committee's warning against single-eyewitness convictions — show it changed practice. However, students with limited driving experience and the absence of real emotional strain mean the size of the effect in real life may differ, and the study is reductionist in considering only one variable.”